Yu Sugihara

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Education

2023	**Ph. D** Graduate School of Agriculture, Kyoto University, Japan Advisor: Prof. Ryohei Terauchi
2020	**MSc** Graduate School of Agriculture, Kyoto University, Japan Advisor: Prof. Ryohei Terauchi
2018	**BSc** Department of Agriculture, Kyoto University, Japan

Research Experience

Jan. 2025 – present	Team Leader at The Sainsbury Laboratory, UK
Apr. 2024 – Nov. 2024	Postdoc at Iwate Biotechnology Research Center, Japan
Apr. 2023 – Mar. 2024	Postdoc at The Sainsbury Laboratory, UK (Kamoun group)
Oct. 2021 – Mar. 2023	Internship at The Sainsbury Laboratory, UK (Kamoun group)
Apr. 2019 – Sep. 2021	Visiting Researcher at Iwate Biotechnology Research Center, Japan

Publications

1. Toghani A, Frijters R, Bozkurt TO, Terauchi R, Kamoun S*, **Sugihara Y*** (2025) Can AI modelling of protein structures distinguish between sensor and helper NLR immune receptors? *New Phytologist*, 248: 17–23. DOI: 10.1111/nph.70391 *Corresponding authors

2. **Sugihara Y**, Kourelis J, Contreras MP, Pai H, Harant A, Selvaraj M, Toghani A, Martinez-Anaya C*, Kamoun S* (2025) Helper NLR immune protein NRC3 evolved to evade inhibition by a cyst nematode virulence effector. *PLOS Genetics*, 21: e1011653. DOI:10.1371/journal.pgen.1011653 *Corresponding authors
3. Selvaraj M, Toghani A, Pai H, **Sugihara Y**, Kourelis J, Yuen ELH, Ibrahim T, Zhao H, Xie R, Maqbool A, Concepcion JCD la, Banfield MJ, Derevnina L, Petre B, Lawson DM, Bozkurt TO, Wu C-H, Kamoun S*, Contreras MP* (2024) Activation of plant immunity through conversion of a helper NLR homodimer into a resistosome. *PLOS Biology*, 22: e3002868. DOI:10.1371/journal.pbio.3002868 *Corresponding authors
4. Huang C-Y[+], Huang Y-S[+], **Sugihara Y**, Wang H-Y, Huang L-T, Lopez-Agudelo JC, Chen Y-F, Lin K-Y, Chiang B-J, Toghani A, Kourelis J, Wang C-H, Derevnina L, Wu C-H (2024) Subfunctionalization of NRC3 altered the genetic structure of the *Nicotiana* NRC network. *PLOS Genetics* 20: e1011402. DOI:10.1371/journal.pgen.1011402 [+]Authors contributed equally
5. Sugimura Y, Oikawa K, **Sugihara Y**, Utsushi H, Kanzaki E, Ito K, Ogasawara Y, Fujioka T, Takagi H, Shimizu M, Shimono H, Terauchi R, Abe A (2024) Impact of rice *GENERAL REGULATORY FACTOR14h (GF14h)* on low-temperature seed germination and its application to breeding. *PLOS Genetics*, 20: e1011369. DOI: 10.1371/journal.pgen.1011369
6. Barragan AC*[+], Latorre SM[+], Malmgren A, Harant A, Win J, **Sugihara Y**, Burbano HA, Kamoun S*, Langner T* (2024) Multiple horizontal mini-chromosome transfers drive genome evolution of clonal blast fungus lineages. *Molecular Biology and Evolution,* 41: msae164. DOI: 10.1093/molbev/msae164 [+]Authors contributed equally, *Corresponding authors
7. **Sugihara Y**, Abe Y, Takagi H, Abe A, Shimizu M, Ito K, Kanzaki E, Oikawa K, Kourelis J, Langner T, Win J, Białas A, Lüdke D, Contreras MP, Chuma I, Saitoh H, Kobayashi M, Zheng S, Tosa Y, Banfield MJ, Kamoun S*, Terauchi R*, Fujisaki K* (2023) Disentangling the complex gene interaction networks between rice and the blast fungus identifies a new pathogen effector. *PLOS Biology*, 21: e3001945. DOI: 10.1371/journal.pbio.3001945 *Corresponding authors
8. Nishiyama C, Sekiguchi S, **Sugihara Y**, Nishikawa M, Makita N, Segawa T, Terasaki M, Takagi H, Koyanagi T (2023) Phylogenomic analysis in *Latilactobacillus sakei* by using polymorphisms detected by next-generation sequencing. *Bioscience of Microbiota, Food and Health*, advpub: 2022–017. DOI: 10.12938/bmfh.2022-017
9. Saiga S, Tada M, Segawa T, **Sugihara Y**, Nishikawa M, Makita N, Sakamoto M, Tanaka K, Wada T, Takagi H (2023) NGS-based genome wide association study helps to develop co-dominant marker for the physical map-based locus of PFRU controlling flowering in cultivated octoploid strawberry. *Euphytica*, 219: 6. DOI: 10.1007/s10681-022-03132-7
10. Takeda T*, Takahashi M, Shimizu M, **Sugihara Y,** Saitoh H, Fujisaki K, Ishikawa K, Utsushi H, Kanzaki E, Sakamoto Y, Abe A, Terauchi R* (2022) Rice apoplastic CBM1-interacting protein

counters blast pathogen invasion by binding conserved carbohydrate binding module 1 motif of fungal proteins. *PLOS Pathogens* 18: e1010792. DOI: 10.1371/journal.ppat.1010792 *Corresponding authors

11. Natsume S, **Sugihara Y**, Kudoh A, Oikawa K, Shimizu M, Ishikawa Y, Nishihara M, Abe A, Innan H, Terauchi R (2022) Genome Analysis Revives a Forgotten Hybrid Crop Edo-dokoro in the Genus *Dioscorea*. *Plant and Cell Physiology*, pcac109. DOI: 10.1093/pcp/pcac109
12. Shimizu M, Hirabuchi A, **Sugihara Y**, Abe A, Takeda T, Kobayashi M, Hiraka Y, Kanzaki E, Oikawa K, Saitoh H, Langner T, Banfield MJ, Kamoun S, Terauchi R (2022) A genetically linked pair of NLR immune receptors show contrasting patterns of evolution. *Proceedings of the National Academy of Sciences*, 119: e2116896119. DOI: 10.1073/pnas.2116896119
13. **Sugihara Y**, Young L, Yaegashi H, Natsume S, Shea DJ, Takagi H, Booker H, Innan H, Terauchi R, Abe A (2022) High-performance pipeline for MutMap and QTL-seq. *PeerJ*, 10: e13170. DOI: 10.7717/peerj.13170
14. **Sugihara Y***, Kudoh A*, Tamiru-Oli M, Takagi H, Natsume S, Shimizu M, Abe A, Asiedu R, Asfaw A, Adebola P, Terauchi R (2021) Population genomics of yams: evolution and domestication of *Dioscorea* species. In: *Population Genomics*. *Springer*, *Cham*. DOI: 10.1007/13836_2021_94 *Authors contributed equally
15. Segawa T, Nishiyama C, Tamiru-Oli M, **Sugihara Y**, Abe A, Sone H, Itoh N, Asukai M, Uemura A, Oikawa K, Utsushi H, Ikegami-Katayama A, Imamura T, Mori M, Terauchi R, Takagi H (2021) Sat-BSA: an NGS-based method using local de novo assembly of long reads for rapid identification of genomic structural variations associated with agronomic traits. *Breeding Science*, 71: 299–312. DOI: 10.1270/jsbbs.20148
16. **Sugihara Y**, Darkwa K, Yaegashi H, Natsume S, Shimizu M, Abe A, Hirabuchi A, Ito K, Oikawa K, Tamiru-Oli M, Ohta A, Matsumoto R, Agre P, De Koeyer D, Pachakkil B, Yamanaka S, Muranaka S, Takagi H, White B, Asiedu R, Innan H, Asfaw A*, Adebola P*, Terauchi R* (2020) Genome analyses reveal the hybrid origin of the staple food crop white Guinea yam (*Dioscorea rotundata*). *Proceedings of the National Academy of Sciences*, 17: 31987–31992. DOI: 10.1073/pnas.2015830117 *Corresponding authors

Preprints

1. Jeong HM, Sugihara Y, Webster MW, Carella P (2025) Stepwise and lineage-specific divergence of a major immune co-chaperone complex in leptosporangiate ferns. *bioRxiv*. DOI:10.64898/2025.12.17.694876
2. Pai H, Sakai T, Posbeyikian A, Frijters R, Sugihara Y, Contreras MP, Kourelis J, Adachi H, Kamoun S, Toghani A (2025) A hierarchical immune receptor network in lettuce reveals contrasting patterns of evolution in sensor and helper NLRs. *bioRxiv*. DOI:10.1101/2025.02.25.639832

3. Kudoh A, Natsume S, Sugihara Y, Kato H, Abe A, Oikawa K, Shimizu M, Itoh K, Tsujimura M, Takano Y, Sakai T, Adachi H, Ohta A, Ohtsu M, Ishizaki T, Terachi T, Terauchi R (2025) Whole-genome sequencing reveals the molecular basis of sex determination in the dioecious wild yam *Dioscorea tokoro*. *bioRxiv*. DOI:10.1101/2025.04.28.650915
4. Fujisaki K, Abe Y, **Sugihara Y**, Nemoto K, Ito K, Kanzaki E, Ishikawa K, Iwai M, Utsushi H, Saitoh H, Takagi H, Takeda T, Abe A, Zheng S, Bialas A, Banfield MJ, Kamoun S, Terauchi R (2024) Binding of a pathogen effector to rice Exo70 proteins tethered to the NOI/RIN4 integrated domain of the NLR receptor Pii2 confers immunity against fungi. *bioRxiv*. DOI:10.1101/239400

Presentations

1. AmirAli Toghani, Sophien Kamoun, **Sugihara Y**. AI-empowered genome annotation tools for large-scale genomics. 1st Tunisian Bioinformatics Conference, Sousse, Tunisia. October 2025.
2. AmirAli Toghani, Sophien Kamoun, **Sugihara Y**. Deep-learning-based annotation of 346 genomes in the Solanaceae family yields a harmonized dataset of 15 million proteins. SolRub2025 Conference, Turin, Italy. September 2025.
3. **Sugihara Y**, Toghani A, Frijters R, Bozkurt TO, Terauchi R, Kamoun S. Can AI modelling of protein structures distinguish between sensor and helper NLR immune receptors? *2025 IS-MPMI*. Cologne, Germany. July 2025.
4. **Sugihara Y**, Kourelis J, Contreras MP, Pai H, Selvaraj M, Toghani A, Martinez-Anaya C, Kamoun S. Helper NLR immune protein NRC3 evolved to evade inhibition by a cyst nematode virulence effector. *The 26th Annual Meeting of Society for Evolutionary Studies, Japan*. Kanagawa, Japan. August 2024.
5. **Sugihara Y**, Białas A, Langner T, Barragan AC, Kourelis J, Abe Y, Fujisaki K, Banfield MJ, Terauchi R, Kamoun S. The NLR immune receptor Pik-1 evolved to respond to fungal effectors of the AVR-Mgk family early in the evolution of *Oryza* and prior to rice domestication. *2023 IS-MPMI*. Rhode Island, PA. July 2023.
6. **Sugihara Y**. Genetics approach using whole genome sequencing to rapidly identify the genomic loci involved in biological traits of rice blast fungus. *12th Japan-US Seminar in Plant Pathology*. Zoom workshop. October 2020.
7. **Sugihara Y**, Natsume S, Abe A, Shimizu M, Obidiegwu J, Terauchi R. Population genomics of *Dioscorea tokoro*, a wild yam species. *Plant and Animal Genome (PAG) XXVII Conference*. San Diego, CA. January 2019.
8. **Sugihara Y,** Abe A, Shimizu M, Yaegashi H, Natsume S, Obidiegwu J, Terauchi R. Application of population genomics and GWAS to *Dioscorea rotundata* (white Guinea yams) for identifying loci controlling agronomically important traits. *Plant and Animal Genome (PAG) XXVII Conference*. San Diego, CA. January 2019.

Workshops

1. GetGenome Team. Symposium on Bridging AI and Biology: The TSL-KCGEB Joint Exploration of AI and Plant Pathogen Systems. KCGEB-TSL Joint Workshop. May 2025.

Awards

1. Postdoctoral oral presentation award, *The 26th Annual Meeting of Society for Evolutionary Studies, Japan*. Aug. 2024

Peer reviews

- New Phytologist (3)
- BMC Biology (2)
- Genetics (2)
- eLife (2)
- Molecular Biology and Evolution (1)
- Ecology and Evolution (1)
- Rice (1)
- Microbial genomics (1)